Ex

Sec 4.3

Solve the system

$$\begin{cases} 4x + y - 3z = 11 \\ 2x - 3y + 2z = 9 \\ x + y + z = -3 \end{cases}$$

I notice that in eq 3 I have an x and eq(3) I have a $4x$ so If I multiply eq (3) by -4 and add to eq (1) I will cancel the x's

So

$-4(\text{eq } 3) \Rightarrow -4(x + y + z) = -4(-3)$

$= -4x - 4y - 4z = 12$

$-4(\text{eq } 3) + \text{eq } 1 \Rightarrow$

$$\begin{array}{r} -4x - 4y - 4z = 12 \\ +\ 4x + y - 3z = 11 \\ \hline 0x - 3y - 7z = 23 \end{array}$$

Now I take this new equation and replace either eq (1) or eq(3) I'll choose eq(3) because I like the coefficient's of 1 in eq(3)

So we have the new but still equivlent System

$$\begin{cases} -3y - 7z = 23 & \text{eq 1} \\ 2x - 3y + 2z = 9 & \text{eq 2} \\ x + y + z = -3 & \text{eq 3} \end{cases}$$

NEXT $-1(\text{eq 1}) \Rightarrow -1(-3y - 7z) = -1 \cdot 23$

$= 3y + 7z = -23$ and add to eq(2)

$$\begin{array}{r} 3y + 7z = -23 \\ +2x - 3y + 2z = 9 \\ \hline 2x + 0y + 9z = -14 \end{array}$$

I'll replace eq ②

New System:

$$\begin{cases} -3y - 7z = 23 \\ 2x + 9z = -14 \\ x + y + z = -3 \end{cases}$$

Hmm at this point any more operations and i'll pick up a variable I already removed this is ok and will happen so we just continue and get rid of what we pick up later.

I'll now do $-2(eq\,3) + (eq\,2)$

$$\begin{array}{rll} -2\ eq(3) = & -2x - 2y - 2z = 6 \\ + \ eq\,2 & +2x \quad\quad + 9z = -14 \\ \hline & 0x - 2y + 7z = -8 \end{array}$$

Replacing $eq(2)$ Gives

$$\begin{cases} -3y - 7z = 23 \\ -2y + 7z = -8 \\ x + y + z = -3 \end{cases}$$

Now Lets try $3(eq_2) + (-2)(eq\,1)$

$$\begin{array}{rll} 3(eq_2) = & \cancel{-6y} + 21z = -24 \\ + -2(eq\,1) & + \cancel{6y} + 14z = -46 \\ \hline & 0y + 35z = -70 \end{array}$$

Our Goal will be to take that system and put it into what is called Row Echolon form which will allow us to Back Substitute to solve the system.

we will use Gaussian Elimination to turn

This

$$\begin{cases} 4x + y - 3z = 11 \\ 2x - 3y + 2z = 9 \\ x + y + z = -3 \end{cases}$$

To →

Row Echelon form

This

$$\begin{cases} x + y + z = -3 \\ -2y + 7z = -8 \\ z = -2 \end{cases}$$

Which is easy to solve $\boxed{z=-2} \Rightarrow y+-2=-5 \Rightarrow \boxed{y=-3} \Rightarrow x-3-2=-3 \Rightarrow \boxed{x=2}$

What can we do with Gaussian Elimination

1) Inter change two equations
2) multiply equations by non zero constants
3) Add multiples of two equations togeter and replace one of the two with the resulting equation.

lets try it.

Replacing eq (1) with our new equation gives:

$$\begin{cases} 35z = -70 \\ -2y + 7z = -8 \\ x + y + z = -3 \end{cases}$$

Now I will swap the rows to get Echelon form

$$\begin{cases} x + y + z = -3 \\ -2y + 7z = -8 \\ 35z = -70 \end{cases}$$

Notice If we do $\frac{1}{35}$(eq3) we get

$\frac{1}{35}(35z) = \frac{1}{35}(-70 \Rightarrow z = -2$

Replacing we get

$$\begin{cases} x + y + z = -3 \\ -2y + 7z = -8 \\ z = -2 \end{cases}$$

Now we can Back Substitute to solve

$\boxed{z = -2}$ $\Rightarrow$ $-2y + 7(-2) = -8$

$-2y - 14 = -8$

$+14 \quad +14$

$\frac{-2y}{-2} = \frac{6}{-2}$

$\boxed{y = -3}$

$z = -2$, $y = -3$ $\Rightarrow$ $x - 3 - 2 = -3$

$+3 \quad +3$

$x - 2 = 0$

$+2 \quad +2$

$\boxed{x = 2}$

So $\boxed{x = 2 \quad y = -3 \quad z = -2}$

$$\begin{cases} x + y - 3z = -1 \\ y - z = 0 \\ 0 = 0 \end{cases}$$ ← when you get 0=0 you

have infinately many solutions since this says $0z = 0$ which is true for any number z.

No Solution

When you get to Echelon form you will have something like.

$$\begin{cases} x - 3y + z = 1 \\ 5y - 4z = 0 \\ 0 = -2 \end{cases}$$ ← This statement is untrue so we can't have a solution.

Now lets do the same problem in class but with diffrent choices for our operations.

"In class with clickers"

Number of solutions.

Just like in the 2 variable case we either have:

- One solution.
- Infinitely many solutions
- No solutions

One Solution

You will solve the system and obtain unique values for x, y, z

Infinately many

When you get your system into Echalon form you will get somthing like,